Expressway and MRA

C inside, E at the edge, traversal zone between them.
MRA still registers to CUCM.
Do not copy a port list from memory.
Signaling up does not prove media.
Source: Cisco Expressway deployment guides.

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Sources and go deeper
Opened 2026-10. A row marked confirm has no link, because that page was not verified.
Cisco. Cisco Expressway Series product support. https://www.cisco.com/c/en/us/support/unified-communications/expressway-series/series.html. last checked 2026-10
Cisco. Mobile and Remote Access through Cisco Expressway, X15.2. https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/expressway/config_guide/X15-2/mra/exwy_b_mra-deployment-guide-x152.html. last checked 2026-10
Protocol and ports. Cisco Expressway IP port usage (X14.3 and X15.0). https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/expressway/config_guide/X15-0/ip-port/exwy_b_cisco-expressway-ip-port-usage-configuration-guide-x15.html. last checked 2026-10
Cisco. Cisco Unified Communications Manager product support. https://www.cisco.com/c/en/us/support/unified-communications/unified-communications-manager-callmanager/series.html. last checked 2026-10
Protocol and ports. CUCM 15 TCP and UDP port usage. https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/cucm/admin/15/systemConfig/cucm_b_system-configuration-guide-15/cucm_m_tcp-and-udp-port-usage-12-0.html. last checked 2026-10
Cisco. Cisco Collaboration Preferred Architectures. https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/uc_system/design/guides/PAdocs.html. last checked 2026-10
Community. Cisco Community, Collaboration knowledge base. https://community.cisco.com/t5/collaboration-knowledge-base/tkb-p/4691-docs-collaboration-voice-video. last checked 2026-10
Wikipedia. Voice over IP (Wikipedia). https://en.wikipedia.org/wiki/Voice_over_IP. last checked 2026-10
Cisco. Collaboration 12 SRND book. confirm. The historical Collaboration 12 SRND path opened as a product listing, not the design book. Use the Preferred Architecture index, which was opened. last checked 2026-10
Cisco. System Configuration Guide for CUCM, Release 15. https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/cucm/admin/15/systemConfig/cucm_b_system-configuration-guide-15.html. last checked 2026-10
Protocol and ports. IANA service names and port numbers. https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml. last checked 2026-10
RFC. RFC 3550, RTP. https://www.rfc-editor.org/rfc/rfc3550. last checked 2026-10
RFC. RFC 8445, ICE. https://www.rfc-editor.org/rfc/rfc8445. last checked 2026-10
Wikipedia. Real-time Transport Protocol (Wikipedia). https://en.wikipedia.org/wiki/Real-time_Transport_Protocol. last checked 2026-10